

Learning Streaming Video Representation via Multitask Training

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01 Why Streaming Is Different



- StreamFormer learns streaming video features for OAD, OVIS, and VQA
 - Multiple granularities include global, temporal, and spatial cues
 - Design targets causal and low-latency inference with KV-cache
 - Unified visual-language alignment supports closed and open sets
- Multitask training improves transfer across tasks
- Streaming focus differentiates from offline pipelines
 - Per-frame processing avoids any future leakage
 - Compact memory maintains long-term context
 - Fine detail is preserved versus coarse frame features

02 Problem and Contributions

Problem

- Streaming understanding requires decisions without future frames
 - Strict causality ensures no lookahead during inference
 - Framewise updates maintain responsiveness under load
 - Compact state tracks long-term temporal context
- Latency and memory must remain stable as streams grow
 - KV-cache enables near-constant per-frame cost
 - Causal masks avoid quadratic temporal growth
 - Design supports real-time deployment targets
- Image-encoder plus LLM stacks lose fine spatiotemporal detail
 - Coarse frame tokens harm precise motion reasoning
 - Spatial fidelity drops under aggressive pooling
 - Causal temporal interaction is missing in the backbone

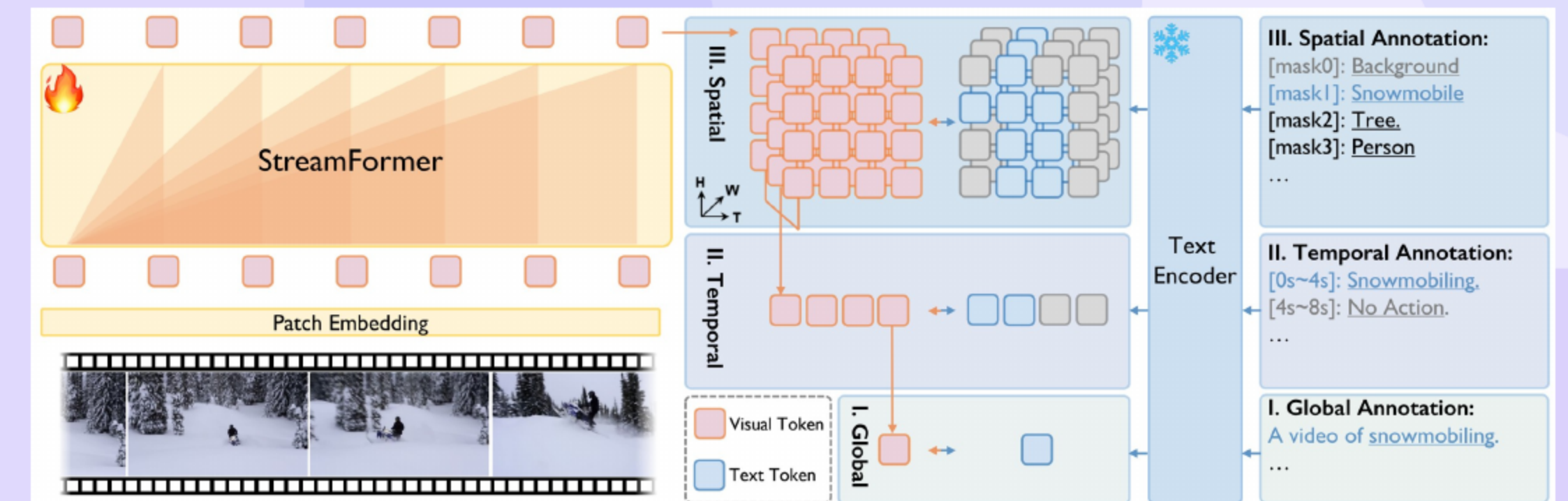
Contributions

- StreamFormer converts an image ViT into a causal streaming backbone
 - Temporal attention becomes strictly causal for streaming
 - Spatial layers inherit **SigLIP initialization** with LoRA
 - Design preserves strong image understanding capability
- Unified multitask visual-language alignment spans three granularities
 - Global, temporal, and spatial features share **text prompts**
 - Closed and open sets are handled with one framework
 - Multitask learning improves cross-task transferability
- Extensive experiments show efficient and strong performance
 - Online action detection improves without optical flow
 - Online VIS surpasses ResNet and SigLIP ViT baselines
 - VideoQA integration matches larger-resolution baselines

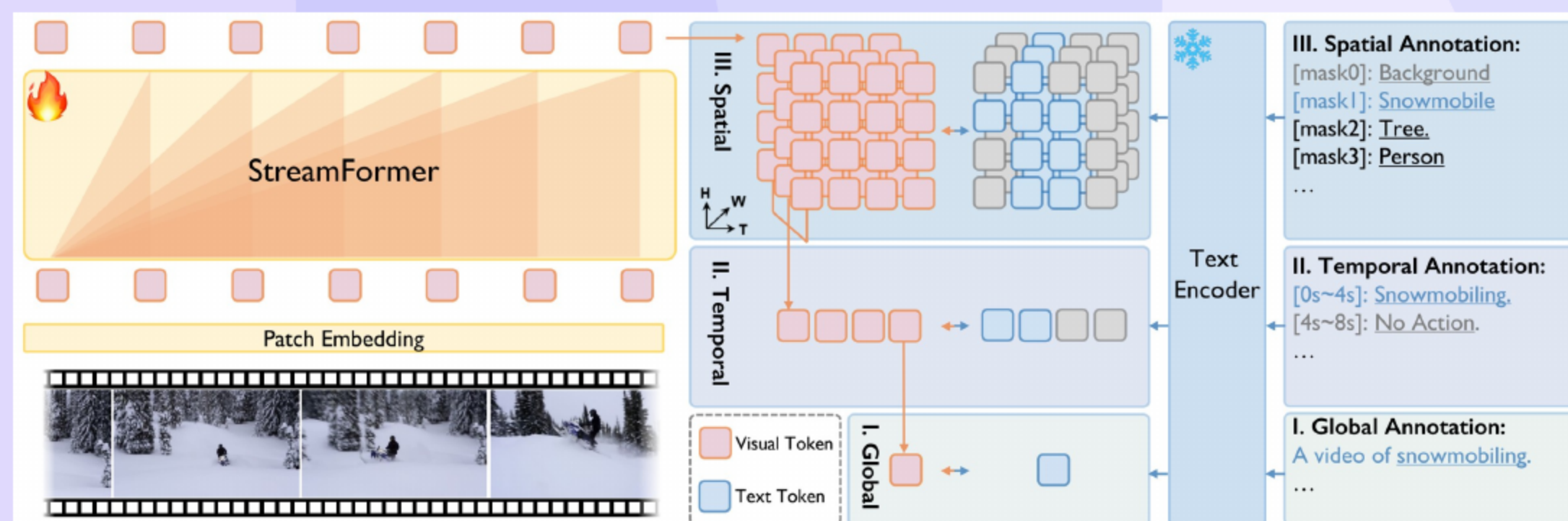
03 Causality and KV-Cache

- Causal attention balances accuracy and efficiency
- KV-cache keeps near-constant latency and memory
- Bidirectional attention raises temporal cost sharply
- Cache reuse stabilizes runtime on long videos
- Causal attention yields online-task gains

- Streaming action understanding leverages adaptive memory and temporal context
 - Historical cues stabilize per-frame predictions
 - Some methods rely on predicted futures or flow
 - Real-time suitability depends on efficient memory
- Video instance segmentation advances emphasize efficiency and openness
 - Unified detection, segmentation, and tracking
 - Open-vocabulary support broadens category coverage
 - Optimized foundations reduce inference overhead
- Language-video pretraining benefits from robust alignment objectives
 - Sigmoid alignment remains strong across batch sizes
 - Masked temporal modeling captures motion regularities
 - Large datasets can be **data-hungry** for training
- Our stance enforces strictly causal processing within the backbone
 - Fine features are preserved for motion reasoning
 - Shared prompts align features across tasks
 - Design avoids reliance on optical flow heuristics



- Divided space-time attention is modified with a temporal causal mask
 - Frames attend only to past and current context
 - Softmax is applied along time under causality
 - Temporal modules are gated to learn gradually
- Spatial attention is initialized from SigLIP and adapted with LoRA
- Backbone outputs support global, temporal, and spatial readouts
 - Patch tensor, frame-wise feature, and last-frame feature are produced
 - All features project into a shared **vision-text space**
 - Shared projections simplify multitask supervision



- Global, temporal, and spatial features align to shared text embeddings
 - Last-frame, per-frame, and per-patch signals are used
 - One alignment interface handles diverse tasks
 - Cross-task transfer emerges from shared prompts
- Closed and open sets are unified through text prompts
 - Category labels convert into descriptive prompts
- Unified alignment reduces task-specific architecture burden
 - Lightweight heads replace bespoke modules
 - Frozen backbone enables efficient adaptation
 - Training scales cleanly across datasets

Objectives

- Global objectives cover action recognition and video-text retrieval
 - Last-frame features preserve strict causality
 - Sigmoid alignment uses learnable temperature and bias
 - Captions provide natural negatives for retrieval
- Temporal objectives address localization and grounding
 - Frame-wise contrastive pairs align events to intervals
 - Batch negatives and background sharpen decisions
 - Per-frame embeddings support robust timing
- Spatial objectives supervise segmentation and referring grounding
 - Patch-text scores upsample into dense masks
 - Cross-entropy and contrastive terms guide pixels
 - Referring inputs enable text-driven masks
- Training alternates tasks under a joint loss
 - Task sampling balances data scale differences

Data Mix

- Pretraining aggregates around one million video-text pairs
 - Datasets cover AR, VTR, TAL, TVG, VOS, and RVOS
 - Public sources provide diverse scenes and actions
 - Annotation quality supports fine-grained learning
- Unified prompts supervise across granularities and sets
 - Global, temporal, and spatial labels are harmonized
 - Shared text space links closed and open vocabularies
 - Consistent prompting eases cross-dataset fusion
- Mixing strategy over-samples underrepresented tasks
 - Balanced sampling avoids domination by large datasets
 - Curriculum effects stabilize early training
 - Task diversity improves **transfer performance**

Training

- Spatial layers initialize from SigLIP with LoRA adaptation
 - Low-rank updates retain **spatial fidelity**
 - Temporal modules are randomly initialized
 - A learnable gate warms up temporal attention
- Training runs with conservative hyperparameters for stability
 - Twenty epochs with batch size 128 and small learning rate
 - Uniform sampling selects sixteen frames per clip
 - Alternating task sampling reduces bias
- Transfer freezes the backbone and trains lightweight heads
 - Frozen features speed up downstream tuning
 - Task-specific heads remain compact
 - Deployment benefits from **efficient adaptation**

OAD Results

- StreamFormer replaces TSN features within an online detector
 - Optional optical flow concatenation remains supported
 - Per-frame mean average precision is reported
 - THUMOS-14 provides the evaluation benchmark
- Accuracy improves substantially without optical flow input
 - Performance surpasses SigLIP image features
 - Results approach early flow-based methods
 - Streaming inference avoids **expensive flow** computation
- Causal features enhance early and stable action predictions
 - Hand-object cues shift labels earlier under causality
 - Temporal smoothing reduces prediction flicker
 - Robustness holds under camera motion and lighting changes

OAD Qualitative 1

- A reaching motion quickly resolves into opening a cabinet
 - Early frames show ambiguity between background and reaching
 - Subtle hand-handle contact raises opening confidence
 - Predictions remain stable without any future access
- Causal attention aggregates short-horizon motion cues
 - Pose and proximity signals are integrated over time
 - Motion continuity strengthens the correct hypothesis
 - Temporal accumulation yields **earlier certainty**
- Label flips diminish as decisive evidence appears
 - Initial uncertainty reflects limited early cues
 - Confidence converges with consistent observations
 - Stream-aligned features prevent over-smoothing

OAD Qualitative 2

- Basketball dribbling remains consistent under camera pan
 - Rapid background motion challenges frame-only models
 - Lighting shifts do not derail predictions
 - Dribble label persists across frames
- RGB-only features capture periodic hand-ball contacts
 - Kinematic patterns substitute for optical flow
 - Causal attention smooths transient noise
 - Temporal structure maintains action identity
- Eliminating flow reduces compute and failure modes
 - No dependency on **flow estimation errors**
 - Lower latency improves real-time viability
 - System remains simpler to deploy

- Fine hand-object interactions tracked causally
 - Finger pose and contact cues drive labels
 - Small motions accumulate across recent frames
 - Stable labels without optical flow
- Example: fastening a watch clasp online
 - Early ambiguity resolves once clasp engages
 - Confidence rises before full closure
 - No future frames are used

Method	Res.	Backbone	COCO-init.	YTVIS19	
				AP	AR ₁
CrossVIS [115]	360p	ResNet50	✓	33.7	35.4
MinVIS [44]	360p	ResNet50	✓	47.4	45.7
IDOL [105]	360p	ResNet50	✓	49.5	47.7
DVIS++ [123]	480p	ResNet50	✓	55.5	51.1
CTVIS [116]	360p	ResNet50	✓	55.1	51.9
CTVIS [116]	224p	ResNet50	✗	43.6	42.7
CTVIS [116]	224p	SigLIP ViT	✗	40.9	41.8
CTVIS [116]	224p	StreamFormer	✗	45.1	43.6

- Frozen StreamFormer + adapter in CTVIS
- Surpasses ResNet50 and SigLIP at 224p
- Streaming-friendly, strictly causal backbone

OVIS Qualitative 1

- A moving dog mask refines as more context accumulates
 - Initial boundaries appear coarse on entry
 - Fine details emerge around ears and legs
 - Stable masks persist after sufficient evidence
- Patch-level alignment preserves fine spatial structure
 - Causal temporal links enforce smooth evolution
 - Identity remains consistent across frames
 - Detail improves without high-resolution inputs
- Streaming constraints are met during refinement
 - No future frames are required for stability
 - Cache enables smooth updates per frame
 - Online tracking maintains **instance identity**

OVIS Qualitative 2

- A cyclist's mask resumes correctly after full occlusion
 - Foreground object blocks the target for several frames
 - Reappearance preserves the same tracking identity
 - Shape continuity is recovered after occlusion
- Feature memory maintains appearance and motion priors
 - Keys and values persist through the occluded gap
 - Per-instance signals guide re-identification
 - Causal updates avoid **future dependence**
- Online operation handles transient visual disruptions
 - Tracking stability improves segmentation quality
 - Identity swaps become rarer under coherence
 - Temporal design resists brief disturbances

- Fast movers maintain masks and IDs
 - Motion blur handled via temporal cues
 - Boundaries refine after quick accelerations
 - Identity persists through rapid motion
- Example: sprinter and baton exchange
 - Short occlusion during handoff recovered
 - Masks realign on reappearance
 - No lookahead during tracking

- StreamFormer integrates into a language-vision assistant
 - Backbone remains frozen during integration
 - Vicuna-based model provides language reasoning
 - Resolution is fixed to 224 by 224 pixels
- Evaluation reports strong results at modest resolution
 - Scores exceed other 224 methods on key benchmarks
 - Results match larger-resolution baselines on MLVU
 - Causal features support temporal question answering
- Setup reuses existing image-text and video-text data
 - Training follows standard LLaVA-Next protocols
 - Video subset adds temporal supervision signals
 - Frozen vision reduces **training overhead**

VideoQA Qualitative 1

- The model answers what occurs after tying shoes
 - Tying is detected in earlier frames under causality
 - Subsequent frames align with running prompts
 - Answer confirms that the person starts running
- Framewise embeddings support stepwise reasoning
 - Events are grounded to specific time spans
 - Question-derived prompts guide retrieval
 - Temporal alignment improves **answer accuracy**
- Causal processing avoids leakage from future frames
 - Reasoning respects streaming constraints
 - Confidence increases as evidence accumulates
 - Prediction remains stable after decision

VideoQA Qualitative 2

- The model counts flips before a gymnast lands
 - Inversion and rotation phases are identified
 - Count completes once landing is detected
 - Answer reports three flips before landing
- Causal accumulation enables reliable counting
 - Per-frame cues aggregate into event counts
 - Before and after relations are preserved
 - Frame evidence reduces **off-by-one** errors
- Temporal readouts align question and visual events
 - Prompt matching anchors segments precisely
 - Confidences stabilize with consistent motion
 - Temporal alignment guides final selection

VideoQA Qualitative 1–2

- The model distinguishes putting on from taking off a jacket
 - Early frames show similar poses for both options
 - Inward sleeve motion increases putting-on confidence
 - Zipper and shoulder cues confirm the correct label
- Causal evidence shifts probabilities decisively over time
 - Hand trajectory and contact direction drive updates
 - Garment deformation provides discriminative signals
 - Confidence rises to a stable **high value**
- Reasoning remains valid without future observations
 - Decisions rely only on seen frames
 - Ambiguity diminishes with accumulating cues
 - Answer remains stable after resolution

VideoQA Qualitative 3–4

- The model decides between entering and leaving a room
 - Early stance appears neutral near the doorway
 - Handle motion and torso direction favor entering
 - Foot placement and gaze consolidate the decision
- Cues accumulate under a causal constraint
 - Door rotation relative to hinge informs direction
 - Center-of-mass trajectory crosses the threshold
 - Confidence climbs toward a **clear answer**
- Inference avoids any future frame dependence
 - Streaming integrity is maintained throughout
 - Probabilities stabilize as motion continues
 - Final answer states entering the room

- A kitchen clip supports action, segmentation, and question answering
 - Actions progress from picking up to pouring coffee
 - Instance masks remain stable through occlusion
 - Answers reference what and what follows lifting
- Shared causal features enable consistent multi-task behavior
 - Text prompts align semantics across tasks
 - Temporal features maintain event order
 - Spatial detail supports accurate masks
- Unified training improves real-time generalization
 - One backbone handles varied objectives
 - Frozen features ease deployment changes
 - Multitask signals yield **robust transfer**

- Pretraining tasks collectively improve online action and QA
 - Global and temporal tasks benefit OAD and VideoQA most
 - Full set delivers the strongest overall metrics
 - Segmentation gains appear with added spatial supervision
- Causal temporal attention outperforms no-temporal baselines
 - Accuracy rises without bidirectional overhead
 - Streaming constraints remain satisfied
 - Temporal modeling reduces **prediction flicker**
- Curated multitask data beats web-scale contrastive with less compute
 - Annotated sets yield higher downstream accuracy
 - Smaller corpora train faster and transfer better
 - Data efficiency provides **stronger returns**

Limitations

- Causality can delay decisions under early ambiguity
 - Initial labels may flip before decisive cues
 - Short evidence windows limit early certainty
 - Delayed transitions reduce **responsiveness**
- Long horizons and viewpoint changes may cause drift
 - Compact state underfits slow context shifts
 - Scene cuts and shakes induce brief errors
 - Recovery occurs after additional context
- Resolution limits and occlusions challenge fine structures
 - Small or thin objects degrade at 224 resolution
 - Prolonged occlusions risk missed re-entry
 - Prompt under-specification reduces answer quality
- Efficiency trades memory for width and head counts
 - KV-cache grows with attention dimensions
 - Mobile targets need pruning or quantization
 - Resource ceilings introduce **latency spikes**

Pointer to Slide 20

- See next slide for details

- Curated multitask beats web-scale contrastive
- Fewer, richer labels improve streaming transfer
- Better data efficiency with lower compute
- Vignette: earlier action disambiguation in OAD

- StreamFormer combines SigLIP initialization with causal temporal attention
 - Spatial LoRA preserves strong image priors
 - Unified prompts align multi-granular features
 - Backbone remains suitable for real-time use
- Multitask training yields strong and efficient online performance
 - Online action detection improves without flow
 - Online VIS competes with stronger baselines
 - VideoQA integration achieves [solid results](#)
- Design choices prioritize deployability under streaming constraints
 - KV-cache stabilizes latency and memory
 - Frozen backbone simplifies adaptation
 - Unified heads reduce engineering overhead

Thank You

Questions & Discussion